

TECHNOLOGIES FOR MOTOR BEHAVIOUR ANALYSIS AND VIRTUAL MODELLING [1]



PRACTICE 4

Cameras - Projections

Laura Cruciani

Role of digital cameras: Convert physical phenomena (light, radiation) into digital images.



- A **digital image** is a matrix of elements called **pixels** (picture elements), where each pixel is associated with a numeric encoding representing a specific concept (intensity, color, depth, or temperature)
- The exact relationship of a digital image to the physical world is determined by the **acquisition process**.
- A key property of digital images is their **grid structure** and the **spatial coherence** of neighboring pixels.

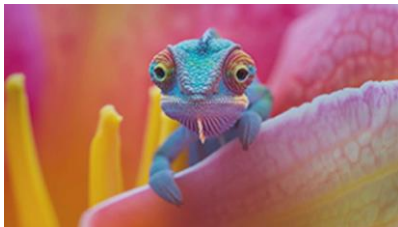
CAMERA



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The type of data stored in an image depends on its category:

Pictorial images (photos): Represent intensity and color information.



Range images: Contain depth information

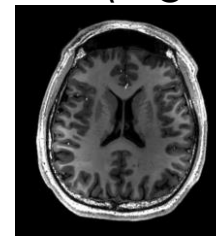


(a) Real-image



(b) Depth-map

Medical images: Encode radiation absorbance levels (e.g., X-ray, CT scans).



Thermal images: Capture heat distribution.



PINHOLE CAMERA MODEL



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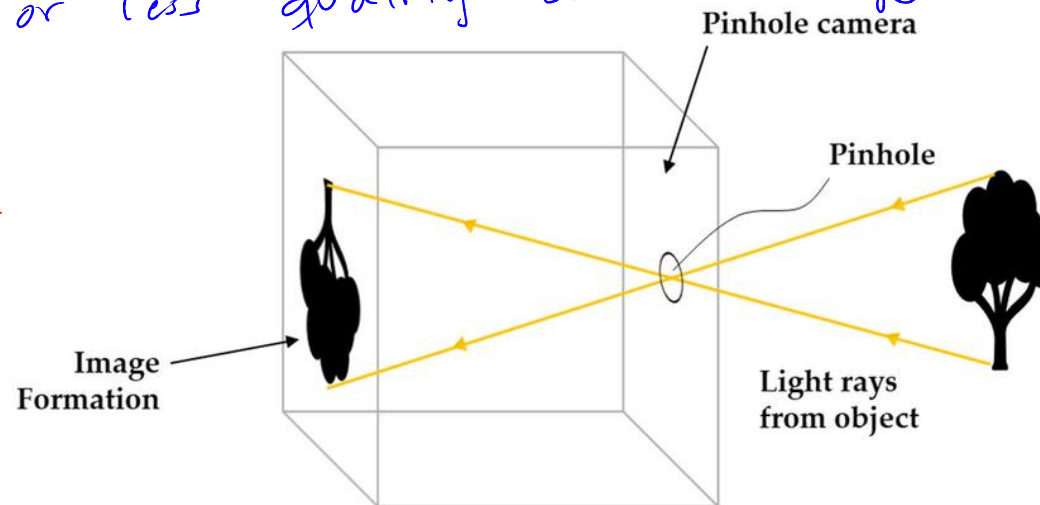
Pinhole Camera: it is a camera with a tiny aperture on one side. The aperture describes how much light passes through the pinhole.

Depending on the amount of light that passes through the pinhole,
there will be more or less quality of the image

it is a trade-off

Pin-hole dimension

Image quality

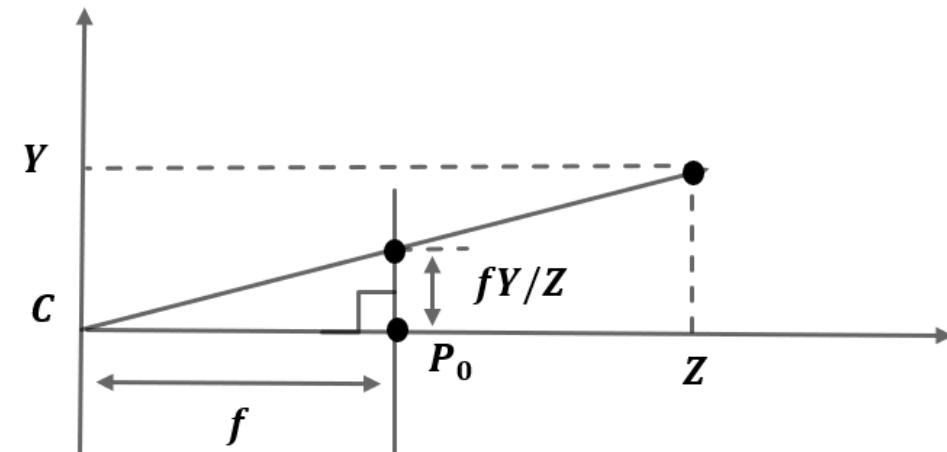
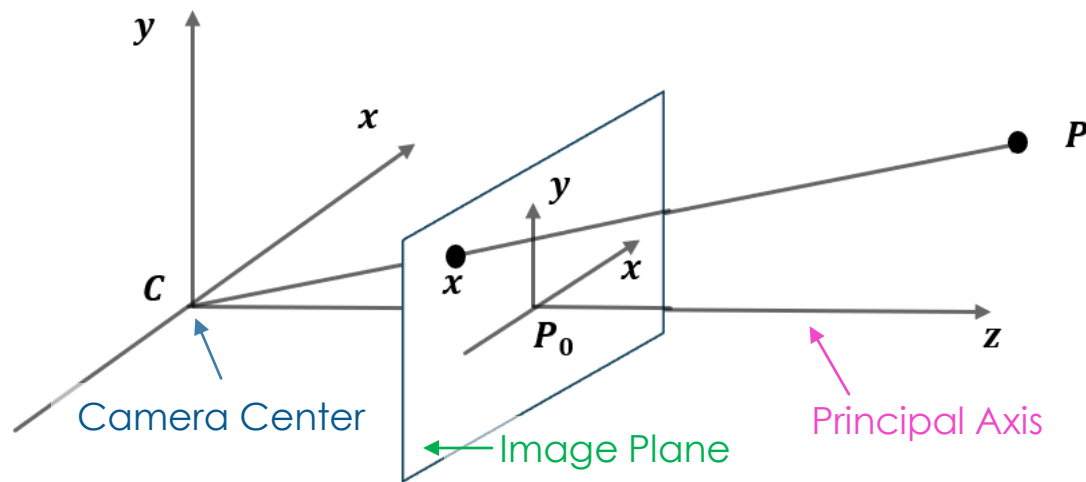


PINHOLE CAMERA MODEL



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Pinhole Camera Model: it describes the mathematical relationship between the coordinates of a point in three-dimensional space and its projection onto the image plane of an ideal pinhole camera, where the camera aperture is described as a point and no lenses are used to focus light.



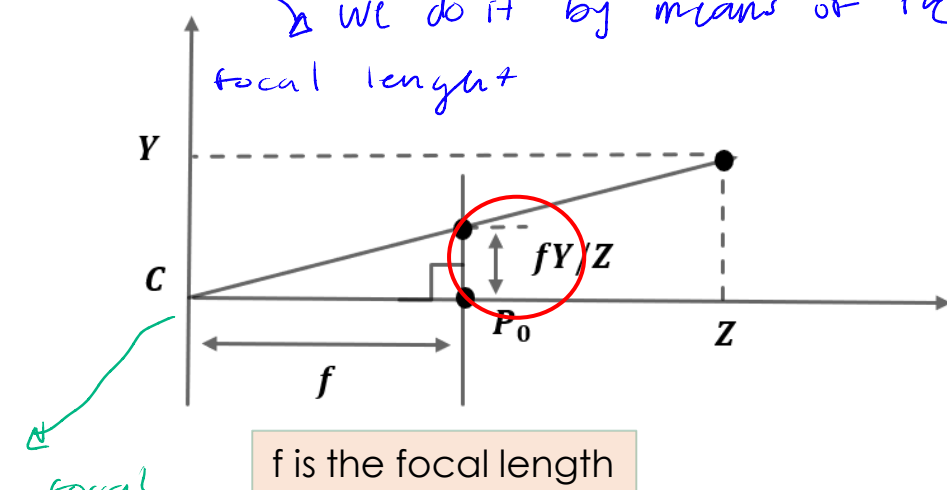
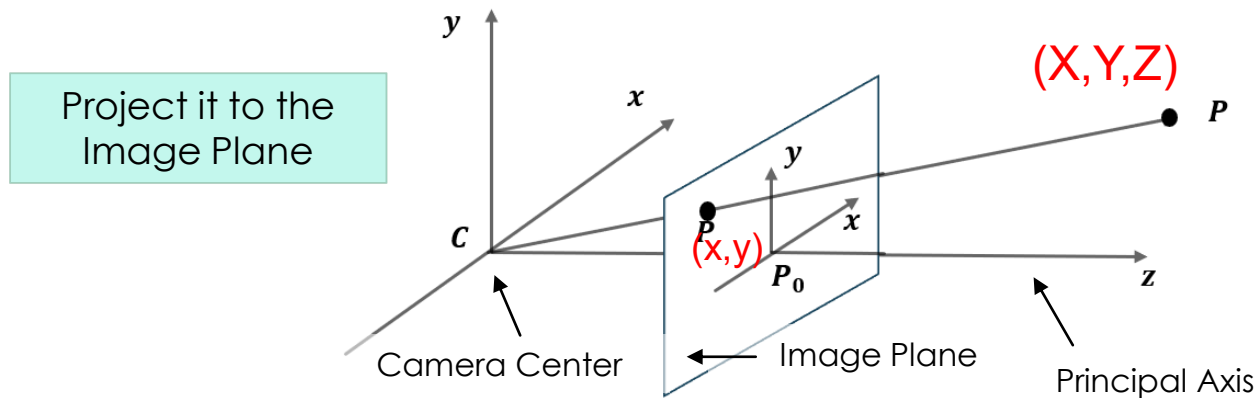
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Pinhole Camera Model:

Our main goal is to go from (X, Y, Z) to its projection to the image plane. We do it by means of the focal length.



$$(X, Y, Z)^T \rightarrow (x, y)^T$$

3D $\leftarrow$ 2D

We can use the focal length and similarity of triangles to find (x, y) .

$$(X, Y, Z)^T \rightarrow \left(\frac{fX}{Z}, \frac{fY}{Z} \right)^T$$

matrix form

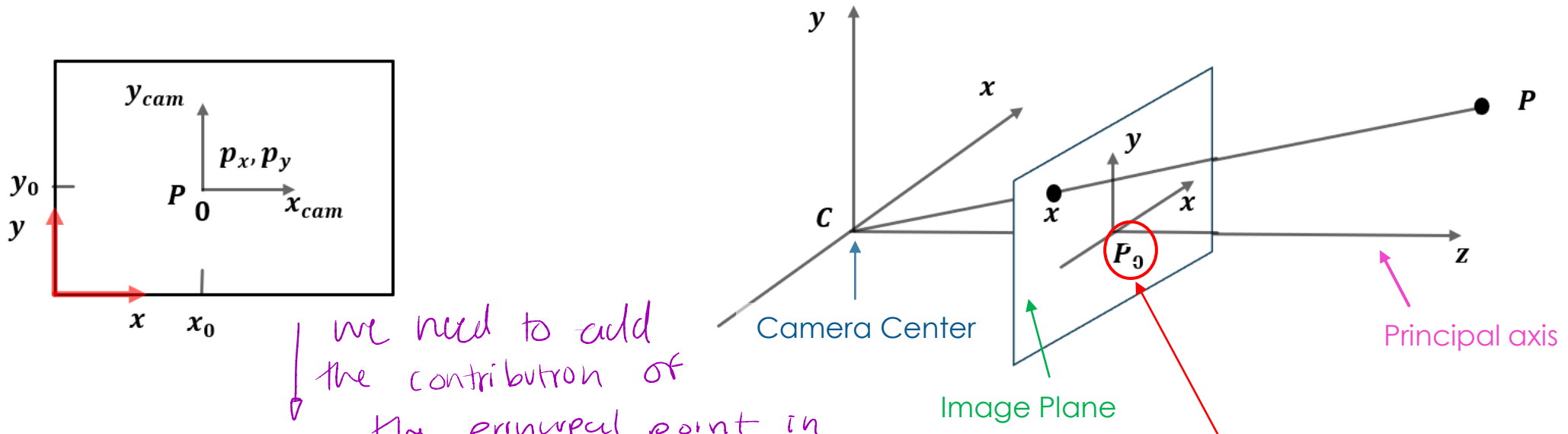
$$\tilde{\mathbf{p}} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

remember we can't go back using the same transformation

PINHOLE CAMERA MODEL



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*we need to add
the contribution of
the principal point in
case it isn't
(0,0)*

$$(X, Y, Z)^T \rightarrow \left(\frac{fX}{Z} + p_x, \frac{fY}{Z} + p_y \right)^T$$

$(p_x, p_y)^T \rightarrow$ coordinates of principal point P_0 on the image plane

Principal Point:
(perpendicular intersection point of
principal axis and image plane)

PINHOLE CAMERA MODEL



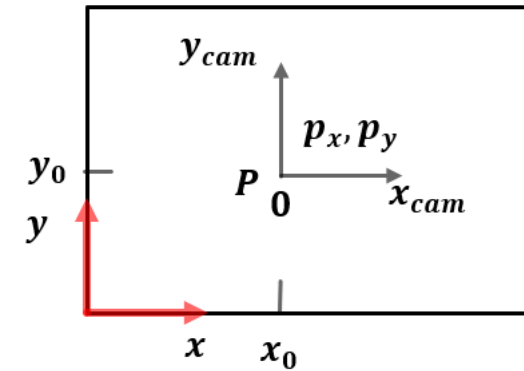
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Matrix form

$$\tilde{\mathbf{p}} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \frac{1}{Z} \begin{bmatrix} f & 0 & p_x & 0 \\ 0 & f & p_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$\rightarrow$ point in the global reference frame.

${}^g\tilde{\mathbf{P}}$



global reference system

$$\begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{K} = \text{Camera Calibration Matrix}$$

it models the intrinsic parameters of the camera

$$\tilde{\mathbf{p}} = \mathbf{K}[\mathbf{I}|\mathbf{0}]\tilde{\mathbf{P}}$$

PINHOLE CAMERA MODEL



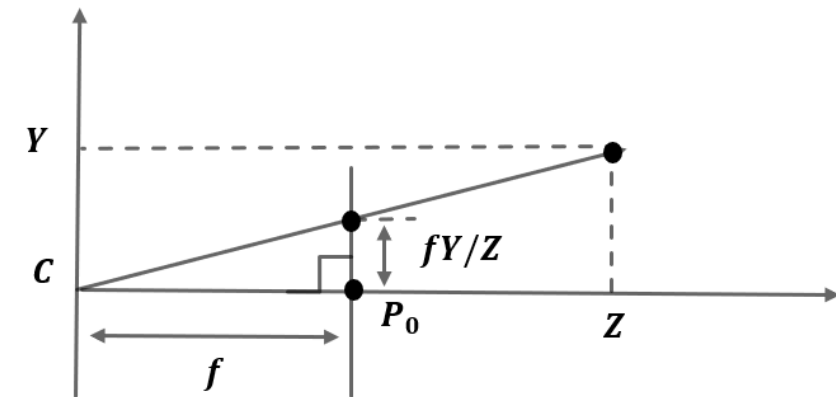
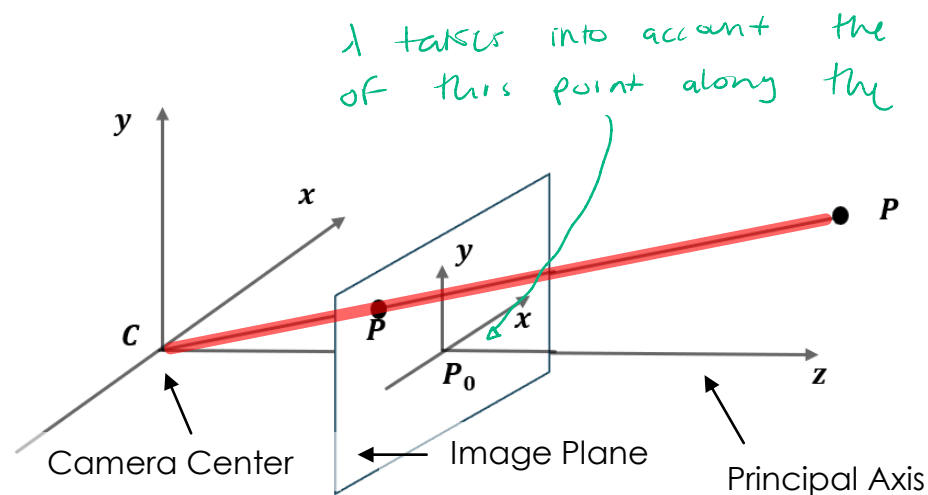
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Projection in compact form:

$$\tilde{\mathbf{p}} = \lambda \mathbf{K} [\mathbf{I} | \mathbf{0}] \tilde{\mathbf{P}}$$

λ can be any real number $\neq 0$

$$[\mathbf{I} | \mathbf{0}] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$



TRANSFORMATIONS

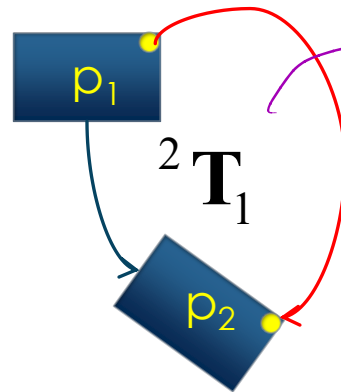
We need to make some transformations between the camera reference plane and global reference plane and these transformations are in terms of rotation and translation



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Transformations: Rotation and Translation of point P_1 leads to point P_2

$$\begin{bmatrix} \mathbf{p}_2 \\ 1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \\ 1 \end{bmatrix} = {}^2\mathbf{T}_1 \begin{bmatrix} \mathbf{p}_1 \\ 1 \end{bmatrix} = {}^2\mathbf{T}_1 \begin{bmatrix} x_1 \\ y_1 \\ z_1 \\ 1 \end{bmatrix}$$



Structure of $\mathbf{T}$

$${}^2\mathbf{T}_1 = \begin{bmatrix} {}^2\mathbf{R}_1 & {}^1\mathbf{P} \\ \mathbf{0}_3 & 1 \end{bmatrix}$$

${}^2\mathbf{R}_1 \rightarrow$ Rotation Matrix

${}^1\mathbf{P} \rightarrow$ Translation vector

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we need it from the camera reference system but it is mostly in general reference system

$$\tilde{\mathbf{p}} = \mathbf{K}[\mathbf{I}|\mathbf{0}]^{camera} \tilde{\mathbf{P}}$$

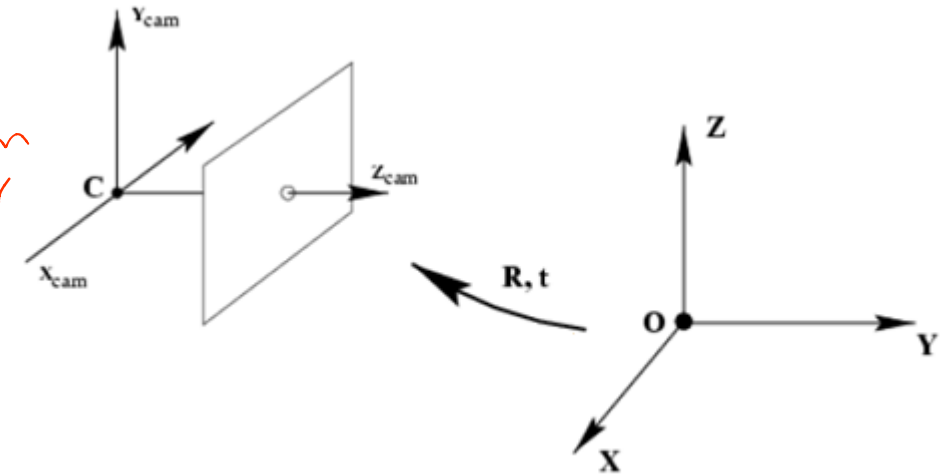
$$^{camera} \tilde{\mathbf{P}} = ^{camera} \mathbf{T}_g \cdot ^g \tilde{\mathbf{P}} = \begin{bmatrix} \mathbf{R}^T & -\mathbf{R}^T \mathbf{C} \\ 0 & 1 \end{bmatrix} ^g \tilde{\mathbf{P}}$$

Translation vector

$$\tilde{\mathbf{p}} = \mathbf{K} \mathbf{R}^T [\mathbf{I} | -\mathbf{C}] ^g \tilde{\mathbf{P}} \quad [\mathbf{I}|\mathbf{0}] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\mathbf{M} = \mathbf{K} \mathbf{R}^T [\mathbf{I} | -\mathbf{C}] = \mathbf{K} \mathbf{H} \rightarrow \tilde{\mathbf{p}} = \mathbf{M} \cdot ^g \tilde{\mathbf{P}}$$

M=new projection matrix (ext. + int.)



PINHOLE CAMERA MODEL

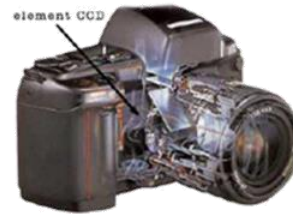


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$$K = \begin{bmatrix} a_x & p_x \\ & a_y & p_y \\ & & 1 \end{bmatrix}$$

intrinsic parameters

4 DOF

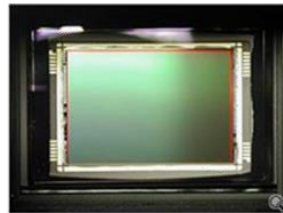


$$M = KR^T [I | -C]$$

Extrinsic parameters

10 DOF

3 from rotation
and 3 from
translation, at the end 10 DOF



$$K = \begin{bmatrix} f & 0 & p_x \\ 0 & f(1+m) & p_y \\ 0 & 0 & 1 \end{bmatrix}$$

Alternative formulation
m: anisotropic scaling factor

PINHOLE CAMERA MODEL



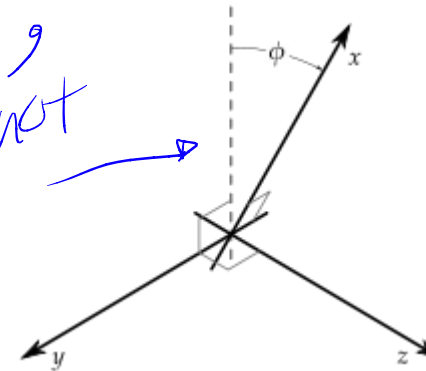
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s skew parameter (Non orthogonal axes)

$$K = \begin{bmatrix} a_x & s & p_x \\ & a_y & p_y \\ & & 1 \end{bmatrix}$$

5 DOF

new parameter, axes are not orthogonal



$$M = KR^T[I \mid -C]$$

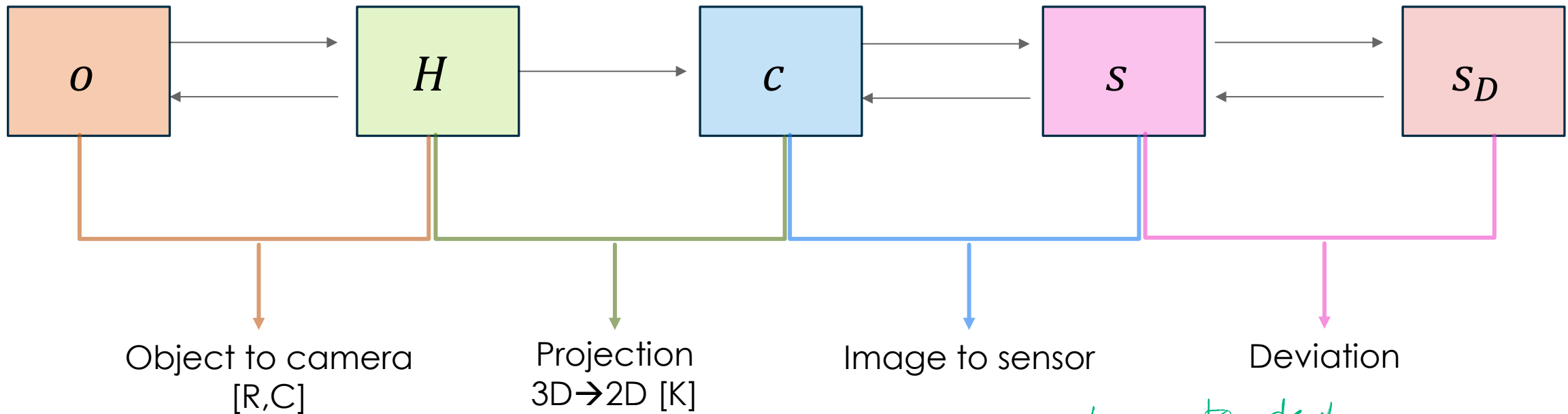
thus 11 DOF

Projective camera with 11 DOF

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By means of translation and rotation we need to go from global frame to camera frame

Then we find the projection and this by means of calibration matrix K

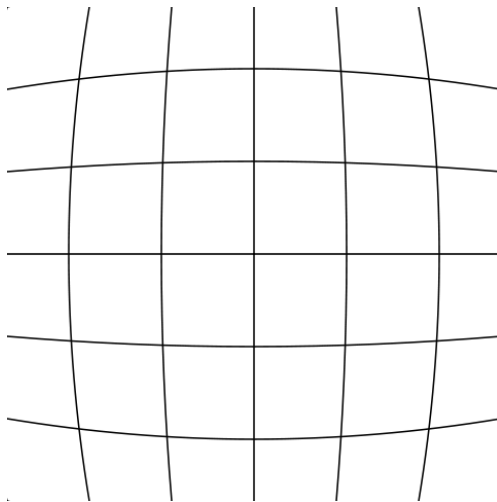
Then we have to deal with deviation or artifacts

DISTORTION

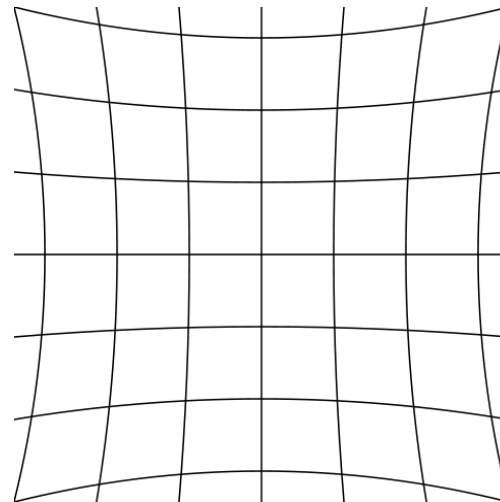


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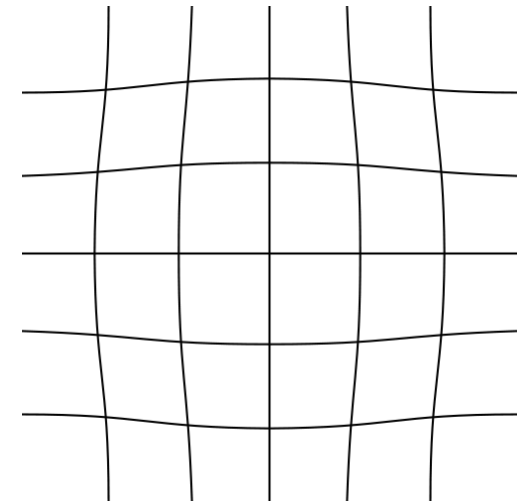
Distortion: Deviation from rectilinear projection, a projection in which straight lines in a scene remain straight in an image. Although distortion can be irregular or follow many patterns, the most commonly encountered distortions are radially symmetric.



Barrel distortion



Pincushion distortion



Mustache distortion

BARREL DISTORTION



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Model of barrel distortion (radial):

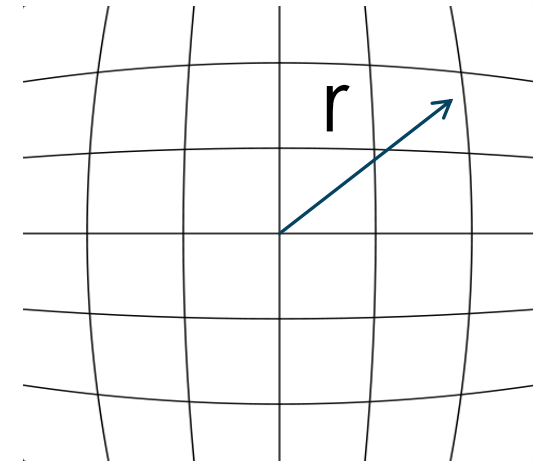
$$\begin{aligned} \overset{\text{Distorted coordinates}}{a_x} &= x(1 + q_1 r^2 + q_2 r^4) \\ \overset{\text{Distorted coordinates}}{a_y} &= y(1 + q_1 r^2 + q_2 r^4) \end{aligned}$$

$\xrightarrow{\text{undistorted coordinates}}$

$[x, y]^T$ Coordinates projected by an ideal camera

r Radial distance from the principal point

q_1, q_2 Model parameters (4th order model)



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straight line distortion



rectified image



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- Add a delta that corrects for the non-linear effects
- Location-dependent shift in the sensor coordinate system
- Individual shift for each pixel
- General mapping:

x, y
Distorted Coordinates

$$\begin{aligned} {}^a x &= {}^s x + \Delta x(\mathbf{p}, \mathbf{q}) \\ {}^a y &= {}^s y + \Delta y(\mathbf{p}, \mathbf{q}) \end{aligned}$$

q
set of distortion parameters
 ${}^s x, {}^s y - [p]$
undistorted coordinates vector

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Camera calibration matrix:

$$K = \begin{bmatrix} f & s & p_x + \Delta x(\mathbf{p}, \mathbf{q}) \\ 0 & f(1+m) & p_y + \Delta y(\mathbf{p}, \mathbf{q}) \\ 0 & 0 & 1 \end{bmatrix}$$

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EXERCISE 1

Camera model is given below. Explain the meaning of the following matrices and the included parameters.

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

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SOLUTION

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

5 DoF

Intrinsic parameters describe the mapping of the scene in front of the camera to the pixels in the final image (sensor).

3x3 rotation and 3x1 translation

Extrinsic parameter denote the coordinate system transformations from 3D world coordinates to 3D camera coordinates (R : Rot and t : trans.)

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SOLUTION

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

f_x and f_y , focal length on x and y axis
(Account for non square pixel using two focal lengths)

s = skew parameter

c_x and c_y principal point coordinates in sensor frame regarding perpendicular intersection point of the optical axis and the image plane.

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SOLUTION

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

X, Y, Z = 3D world coordinates

u, v = pixel coordinates in sensor frame

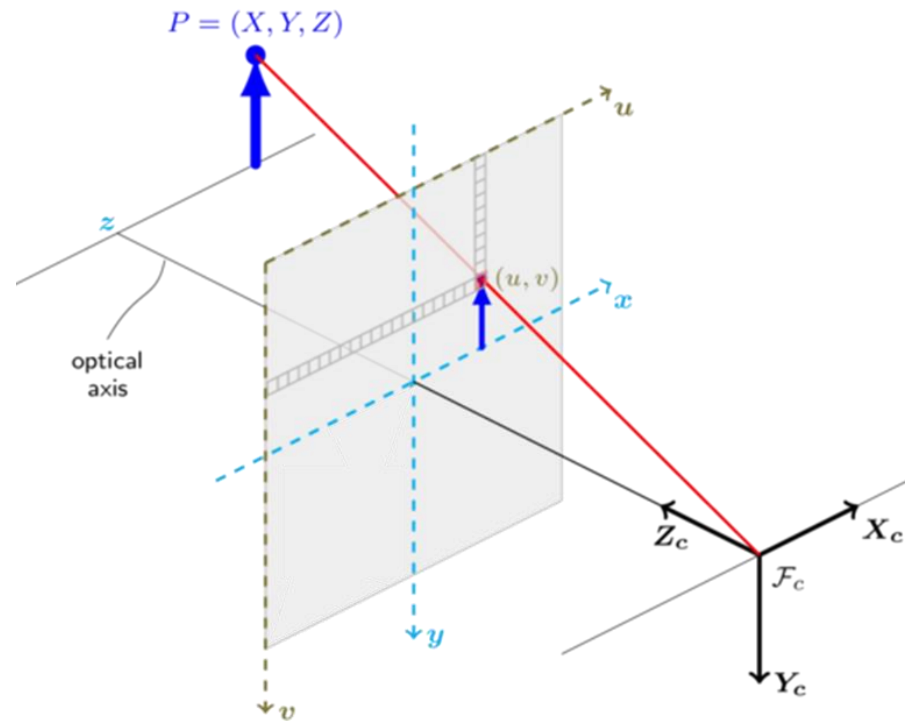
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EXERCISE 2

Show the principal point and focal length on the following figure.

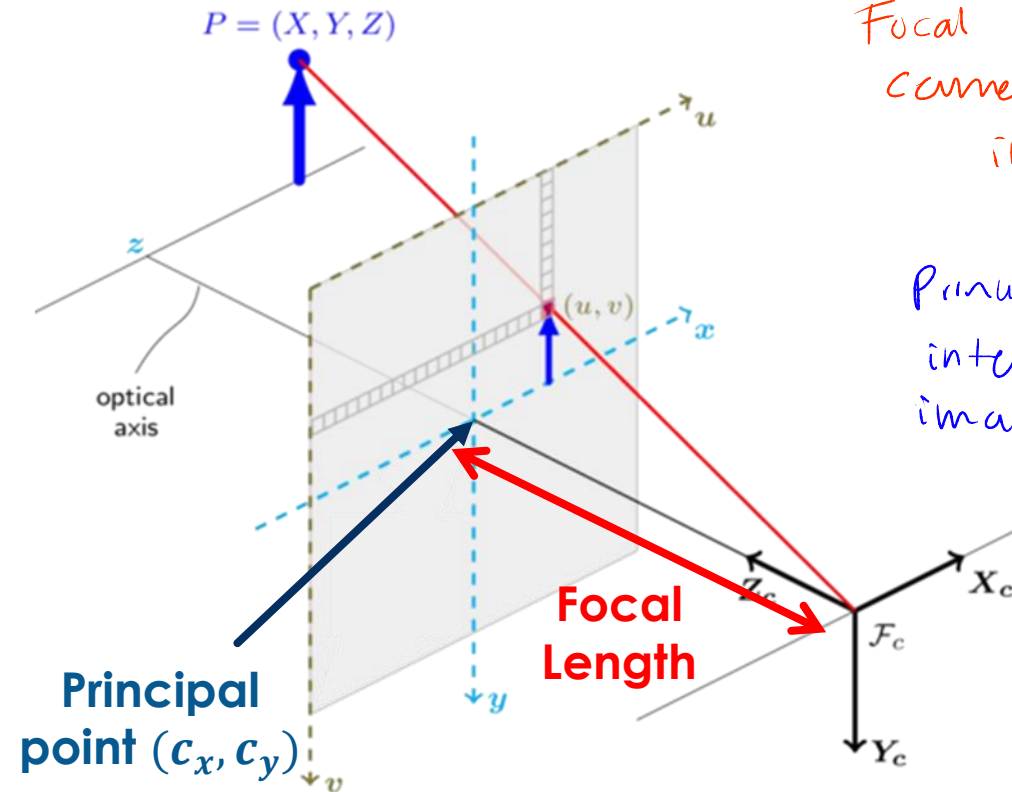


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SOLUTION



Focal length: Distance between camera center and image plane

Principal point: intersection point with the image plane and optical axis

CAMERA MODEL

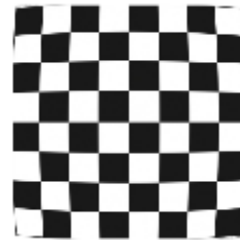


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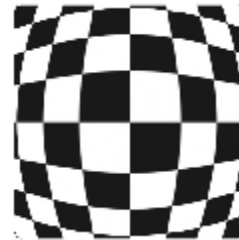
EXERCISE 3

Consider that you have the image of a chess-board. Distortions are reported below. Match each image of figure with the corresponding distortion coefficients:

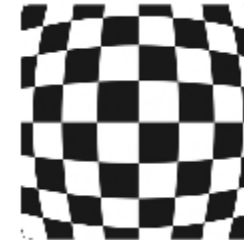
- $q1, q2 = (-0.5, 0.5)$
- $q1, q2 = (-0.1, 0.1)$
- $q1, q2 = (-1, 1)$



(a)

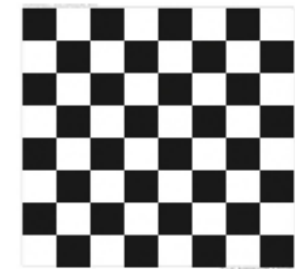


(b)



(c)

Figure Match distortion coefficients



Original Picture

CAMERA MODEL

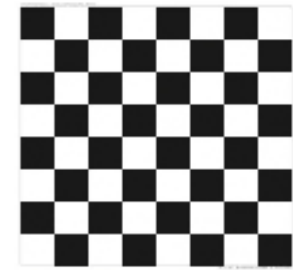


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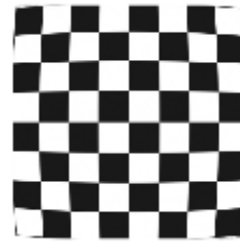
SOLUTION

The higher the distortion coeff, the higher the distortion of the image

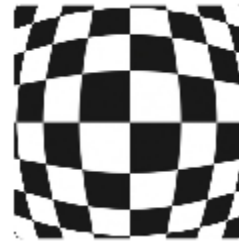
- $q1, q2 = (-0.5, 0.5)$ [c]
- $q1, q2 = (-0.1, 0.1)$ [a]
- $q1, q2 = (-1, 1)$ [b]



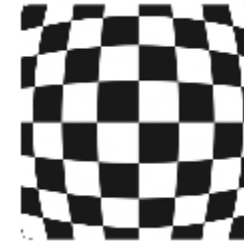
Original Picture



(a)



(b)



(c)

Figure Match distortion coefficients

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EXERCISE 4

- The principal point of a camera has coordinates $^s x_o = 0, ^s y_o = 0$ in the origin of sensor coordinate system. The two distortion coefficients q_1, q_2 are equal to $q_1 = 5 * 10^{-5}$ and $q_2 = 5 * 10^{-8}$.
- Using a 4th order radial distortion model (barrel model), compute the error along x and y axis for the point p, knowing that, ideally, it should lie over the position $(^s x_p, ^s y_p) = (15, 20) \mu m$.

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SOLUTION

- $q_1 = 5 * 10^{-5}$
- $q_2 = 5 * 10^{-8}$
- $x_p = 15\mu m$
- $y_p = 20\mu m$

4° order model:

$${}^a x = x(1 + q_1 r^2 + q_2 r^4)$$

$${}^a y = y(1 + q_1 r^2 + q_2 r^4)$$

$$r = \sqrt{x_p^2 + y_p^2} = 25\mu m \quad \xrightarrow{\text{Radial distance from the principal point}}$$

Handwritten note: $15\mu m^2 + 20\mu m^2$

$$x_{p_a} = x_p * (1 + q_1 * r^2 + q_2 * r^4) = 15.7617 \quad \longrightarrow \quad \Delta_x = x_{p_a} - x_p = 0.7617$$

$$y_{p_a} = y_p * (1 + q_1 * r^2 + q_2 * r^4) = 21.0156 \quad \longrightarrow \quad \Delta_y = y_{p_a} - y_p = 1.0156$$

CAMERA MODEL

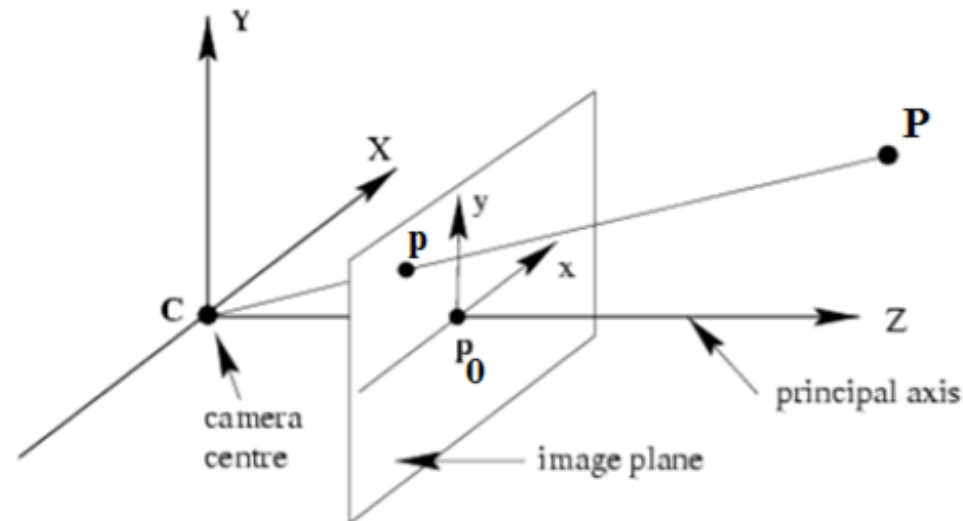


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EXERCISE 5

Suppose to have a projective camera as shown in Figure with an isotropic pixel, skewness is zero and principal point $(0,0)$.

Where Point P is defined as $P(X,Y,Z) = (0, 0.3, 0.4)$ [m]. Where point p is defined as $p(x,y) = (0,3)$ [mm]. Determine the focal length and write the intrinsic matrix by using the computed focal length and information provided at the beginning.



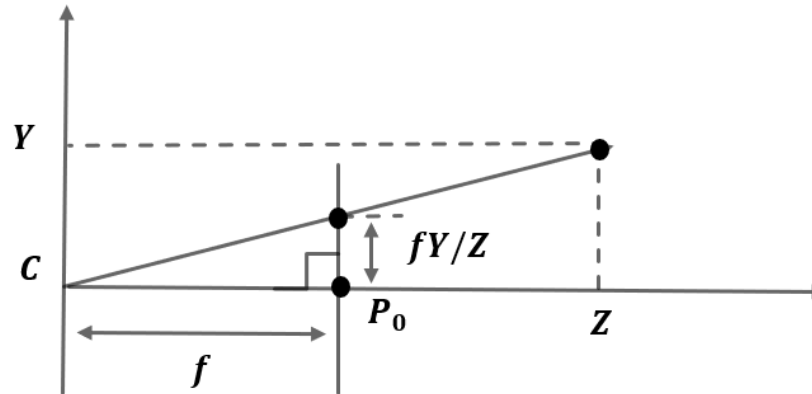
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SOLUTION

- $P(X, Y, Z) = [0, 0.3, 0.4]$
- $p(x, y) = [0.3 * 10^{-3}]$



$$(X, Y, Z)^T \rightarrow \left(\frac{fX}{Z}, \frac{fY}{Z} \right)^T$$

$$y_p = \frac{fY_P}{Z_P} \rightarrow f = \frac{y_p}{Y_P} * Z_P = 4 \text{ [mm]}$$

$$K = \begin{bmatrix} f & s & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

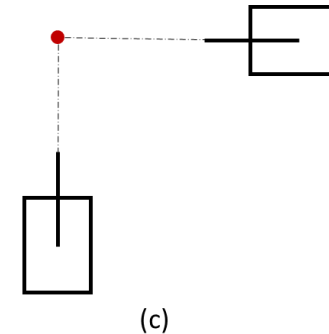
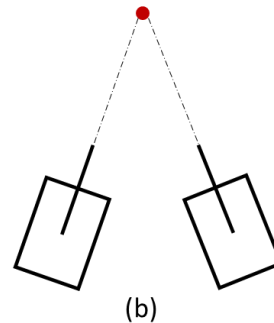
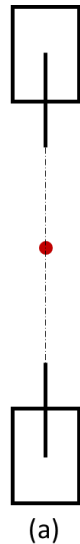
CAMERA MODEL



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EXERCISE 6

In the following figure, three different camera position are given. Determine which one cannot be used for the estimation of a point marker in the space.

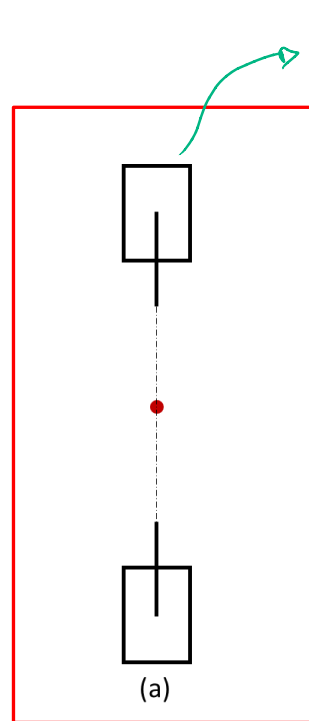


EXERCISE SOLUTION – 6



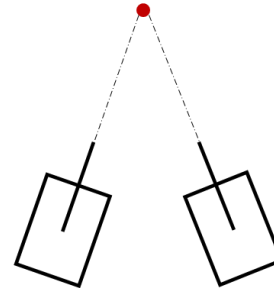
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SOLUTION

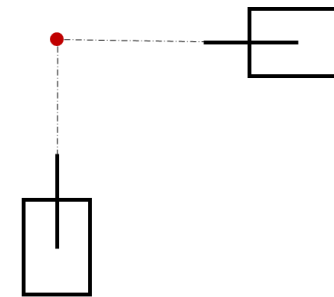


(a)

why?
Because it is not giving
a different perspective of
the point, so we can't estimate the 3D
position of a point



(b)



(c)

CAMERA MODEL

EXERCISE 7

Consider intrinsic and extrinsic camera calibration problem. In the case of two focal lengths present on x and y axis and considering skewness and principal point, how many DoF are required to be found by calibration?

Explain which are those parameters.

CAMERA MODEL



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SOLUTION

Extrinsics $\rightarrow$ 6 DoF

$\rightarrow$ 3 rotation
3 translation

Intrinsics $\rightarrow$ 5 DoF

$\rightarrow$ 11 DoF

- f_x, f_y $\rightarrow$ focal length
- c_x, c_y $\rightarrow$ principal point
- s

CAMERA MODEL

EXERCISE 8

What does the matrix represent?

Define the physical meaning of each element.

$$\blacktriangleright M = \begin{bmatrix} 1 & s & p_x \\ 0 & 1 + m & p_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & 1 \end{bmatrix} R[I|-P_0]$$

CAMERA MODEL

SOLUTION

$$M = \begin{bmatrix} 1 & s & p_x \\ 0 & 1 + m & p_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & 1 \end{bmatrix} R[I] - P_0$$

M : projection matrix of pinhole camera.

c : focal length (distance between optical center and principal point).

s : skew factor.

m : anisotropic scaling factor.

p_x, p_y : principal point offset.

$R[I] - P_0$: extrinsic camera parameters (camera orientation and position in world coordinates).

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EXERCISE 9

What do the equations and their elements represent?

Where they should be put in the matrix of camera calibration?

$$\begin{aligned} {}^a x &= {}^s x + \Delta x(p, q) \\ {}^a y &= {}^s y + \Delta y(p, q) \end{aligned}$$

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SOLUTION

- a_x, a_y : distorted coordinates
- s_x, s_y : undistorted coordinates vector.
- q : set of distortion parameters
- Delta are the actual displacement that the point is subjected to.
- They should be put in the calibration matrix.

$$\begin{aligned} a_x &= s_x + \Delta x(p, q) \\ a_y &= s_y + \Delta y(p, q) \end{aligned}$$

$$K = \begin{bmatrix} f & s & p_x + \Delta x(\mathbf{p}, \mathbf{q}) \\ 0 & f(1+m) & p_y + \Delta y(\mathbf{p}, \mathbf{q}) \\ 0 & 0 & 1 \end{bmatrix}$$

CAMERA MODEL

EXERCISE 10

Consider the following image. One of the images represent a distorted image and the other is the rectified image. Indicate which is distorted and which is rectified.



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SOLUTION



distorted



rectified